

JIAHUI HUANG

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<https://huangjh-pub.github.io/>

WORK EXPERIENCE

Senior Research Scientist, NVIDIA

08/2023 - Now 

- Spatial Intelligence Lab. 3D Computer Vision and Graphics research.
- Building scalable and high-quality 3D reconstruction and data generation algorithms.

Research Intern, NVIDIA

06/2022 - 06/2023

- Toronto AI Lab. Mentor: Prof. Sanja Fidler, Dr. Francis Williams

Research Engineer Intern, Meituan

07/2021 - 08/2021

- Vision Intelligence Center - 3D Mapping Team. Leader: Dr. Zhuoliang Kang
- Deep-learning-based 3D road reconstruction from monocular videos for AV mapping.

EDUCATION

Ph.D., Computer Science, Tsinghua University

09/2018 - 06/2023

- Graphics and Geometry Computing Group. Advisor: Prof. Shi-Min Hu
- PhD Thesis: Dynamic 3D Scene Understanding and Reconstruction.
- Research Focus: 3D Computer Vision, SLAM and Graphics.

Visiting Scholar, Computer Science, Stanford University

02/2020 - 07/2020

- Advisor: Leonidas J. Guibas
- Research Focus: Dynamic Point Clouds Processing and Understanding.

B.S., Computer Science, Tsinghua University

08/2014 - 06/2018

SELECTED PUBLICATIONS

Déjà View: Looping Transformers for Multi-View 3D Reconstruction

arXiv 2026 

Alessandro Burzio, Tobias Fischer, Sven Elflein, Qunjie Zhou, Riccardo de Lutio, Jiawei Ren, Jiahui Huang, Shengyu Huang, Marc Pollefeys, Laura Leal-Taixé, Zan Gojcic, Haithem Turki

Asset Harvester: Extracting 3D Assets from Autonomous Driving Logs for Simulation

arXiv 2026 

Tianshi Cao, Jiawei Ren, Yuxuan Zhang, Jaewoo Seo, Jiahui Huang, Shikhar Solanki, Haotian Zhang, Mingfei Guo, Haithem Turki, Muxingzi Li, Yue Zhu, Sipeng Zhang, Zan Gojcic, Sanja Fidler, Kangxue Yin

TokenGS: Decoupling 3D Gaussian Prediction from Pixels with Learnable Tokens

CVPR 2026 Highlight 

Jiawei Ren, Michal Tyszkiewicz, Jiahui Huang, Zan Gojcic

Lyra 2.0: Explorable Generative 3D Worlds

arXiv 2026 

Tianchang Shen, Sherwin Bahmani, Kai He, Sangeetha Grama Srinivasan, Tianshi Cao, Jiawei Ren, Ruilong Li, Zian Wang, Nicholas Sharp, Zan Gojcic, Sanja Fidler, Jiahui Huang, Huan Ling, Jun Gao, Xuanchi Ren

Depth Completion as Parameter-Efficient Test-Time Adaptation

arXiv 2026 

Bingxin Ke, Qunjie Zhou, Jiahui Huang, Xuanchi Ren, Tianchang Shen, Konrad Schindler, Laura Leal-Taixé, Shengyu Huang

Lyra: Generative 3D Scene Reconstruction via Video Diffusion Model Self-Distillation

ICLR 2026 

Sherwin Bahmani, Tianchang Shen, Jiawei Ren, Jiahui Huang, Yifeng Jiang, Haithem Turki, Andrea Tagliasacchi, David B. Lindell, Zan Gojcic, Sanja Fidler, Huan Ling, Jun Gao, Xuanchi Ren

ViPE: Video Pose Engine for 3D Geometric Perception*arXiv 2025* [🔗](#)

Jiahui Huang, Qunjie Zhou, Hesam Rabeti, Aleksandr Korovko, Huan Ling, Xuanchi Ren, Tianchang Shen, Jun Gao, Dmitry Slepichev, Chen-Hsuan Lin, Jiawei Ren, Kevin Xie, Joydeep Biswas, Laura Leal-Taixé, Sanja Fidler

Gen3C: 3D-Informed World-Consistent Video Generation with Precise Camera Control*CVPR 2025* [🔗](#)

Xuanchi Ren, Tianchang Shen, Jiahui Huang, Huan Ling, Yifan Lu, Merlin Nimier-David, Thomas Müller, Alexander Keller, Sanja Fidler, Jun Gao

Cosmos: World Foundation Model Platform for Physical AI*arXiv 2025* [🔗](#)

NVIDIA Research

VideoPanda: Video Panoramic Diffusion with Multi-View Attention*arXiv 2025* [🔗](#)

Kevin Xie, Amirmojtaba Sabour, Jiahui Huang, Despoina Paschalidou, Greg Klar, Umar Iqbal, Sanja Fidler, Xiaohui Zeng

InfiniCube: Unbounded and Controllable Dynamic 3D Driving Scene Generation with World-Guided Video Models*ICCV 2025* [🔗](#)

Yifan Lu, Xuanchi Ren, Jiawei Yang, Tianchang Shen, Jay Zhangjie Wu, Jun Gao, Yue Wang, Siheng Chen, Mike Chen, Sanja Fidler, Jiahui Huang

Feed-Forward Bullet-Time Reconstruction of Dynamic Scenes from Monocular Videos*NeurIPS 2025* [🔗](#)

Hanxue Liang, Jiawei Ren, Ashkan Mirzaei, Antonio Torralba, Ziwei Liu, Igor Gilitschenski, Sanja Fidler, Cengiz Oztireli, Huan Ling, Zan Gojcic, Jiahui Huang

SCube: Instant Large-Scale Scene Reconstruction using VoxSplats*NeurIPS 2024* [🔗](#)

Xuanchi Ren, Yifan Lu, Hanxue Liang, Jay Zhangjie Wu, Huan Ling, Mike Chen, Sanja Fidler, Francis Williams, Jiahui Huang

STORM: Spatio-Temporal Reconstruction Model for Large-Scale Outdoor Scenes*ICLR 2025* [🔗](#)

Jiawei Yang, Jiahui Huang, Yuxiao Chen, Yan Wang, Boyi Li, Yurong You, Maximilian Igl, Apoorva Sharma, Peter Karkus, Danfei Xu, Boris Ivanovic, Yue Wang, Marco Pavone

fVDB: A Deep-Learning Framework for Sparse, Large Scale, and High-Performance Spatial Intelligence*SIGGRAPH (TOG) 2024* [🔗](#)

Francis Williams, Jiahui Huang, Jonathan Swartz, Gergely Klar, Vijay Thakkar, Matthew Cong, Xuanchi Ren, Ruilong Li, Clement Fuji-Tsang, Sanja Fidler, Eftychios Sifakis, Ken Museth

XCube: Large-Scale 3D Generative Modeling using Sparse Voxel Hierarchies*CVPR 2024 Highlight* [🔗](#)

Xuanchi Ren, Jiahui Huang, Xiaohui Zeng, Ken Museth, Sanja Fidler, Francis Williams

DiffFacto: Controllable Part-Based 3D Point Cloud Generation with Cross Diffusion*ICCV 2023* [🔗](#)

George Kiyohiro Nakayama, Mikaela Angelina Uy, Jiahui Huang, Shi-Min Hu, Ke Li, Leonidas J. Guibas

Neural Kernel Surface Reconstruction*CVPR 2023 Highlight* [🔗](#)

Jiahui Huang, Zan Gojcic, Matan Atzmon, Or Litany, Sanja Fidler, Francis Williams

A Neural Galerkin Solver for Accurate Surface Reconstruction*SIGGRAPH Asia (TOG) 2022* [🔗](#)

Jiahui Huang, Haoxiang Chen, Shi-Min Hu

Multiway Non-rigid Point Cloud Registration via Learned Functional Map Synchronization*T-PAMI 2022* [🔗](#)

Jiahui Huang, Tolga Birdal, Zan Gojcic, Leonidas J. Guibas, Shi-Min Hu

Dynamic 3D Scene Analysis by Point Cloud Accumulation*ECCV 2022* [🔗](#)

Shengyu Huang, Zan Gojcic, Jiahui Huang, Andreas Wieser, Konrad Schindler

MultiBodySync: Multi-Body Segmentation and Motion Estimation via 3D Scan Synchronization	<i>CVPR 2021 Oral</i> 
<u>Jiahui Huang</u> , He Wang, Tolga Birdal, Minhyuk Sung, Federica Arrigoni, Shi-Min Hu, Leonidas J. Guibas.	
DI-Fusion: Online Implicit 3D Reconstruction with Deep Priors	<i>CVPR 2021</i> 
<u>Jiahui Huang</u> , Shi-Sheng Huang, Haoxuan Song, Shi-Min Hu.	
Real-Time Globally Consistent 3D Reconstruction with Semantic Priors	<i>TVCG 2021</i> 
Shi-Sheng Huang, Haoxiang Chen, <u>Jiahui Huang</u> , Hongbo Fu, Shi-Min Hu.	
Subdivision-Based Mesh Convolution Networks	<i>TOG 2021</i> 
Shi-Min Hu, Zheng-Ning Liu, Meng-Hao Guo, Jun-Xiong Cai, <u>Jiahui Huang</u> , Tai-Jiang Mu, Ralph R. Martin.	
ClusterVO: Clustering Moving Instances and Estimating Visual Odometry for Self and Surroundings	<i>CVPR 2020</i> 
<u>Jiahui Huang</u> , Sheng Yang, Tai-Jiang Mu, and Shi-Min Hu.	
WallNet: Reconstructing General Room Layouts from RGB Images	<i>GMOD 2020</i> 
<u>Jiahui Huang</u> , Zhengfei Kuang, Fanglue Zhang, and Shi-Min Hu.	
ClusterSLAM: A SLAM Backend for Simultaneous Rigid Body Clustering and Motion Estimation	<i>ICCV 2019</i> 
<u>Jiahui Huang</u> , Sheng Yang, Zishuo Zhao, Yu-Kun Lai, and Shi-Min Hu.	
DeepPrimitive: Image decomposition by layered primitive detection	<i>CVMJ 2018</i> 
<u>Jiahui Huang</u> , Jun Gao, Vignesh Ganapathi-Subramanian, Hao Su, Yin Liu, Chengcheng Tang, Leonidas J. Guibas.	

ACADEMIC SERVICES

Reviewer , CVPR, ICCV, ICLR, NeurIPS, SIGGRAPH Asia, TVCG	<i>2025</i>
PC Committee/Reviewer , CVPR, ECCV, AAAI, ICML, 3DV, ICLR, CVM	<i>2024</i>
Reviewer , CVPR, ICCV, T-PAMI, NeurIPS, SIGGRAPH, RA-L, TRO, IROS, 3DV, ICLR	<i>2023</i>
Outstanding Reviewer , CVPR 	<i>2022</i>
Reviewer , ECCV, RA-L, IEEE TVCG, IROS, 3DV, Graphics Modeling	<i>2022</i>
Reviewer , ICCV, T-PAMI, NeurIPS, The Visual Computer, IEEE TCSVT	<i>2021</i>
Reviewer , Computer & Graphics, Computational Visual Media	<i>2020</i>
Teaching Assistant , Fundamentals of Computer Graphics, Tsinghua University	<i>02/2019 - 06/2019</i>
Teaching Assistant , Advanced Computer Graphics, Tsinghua University	<i>09/2019 - 01/2020</i>